Data Science & AI

DATA STRUCTURES Through Python

Practice Session

Lecture No.- 01

By- Satya sir

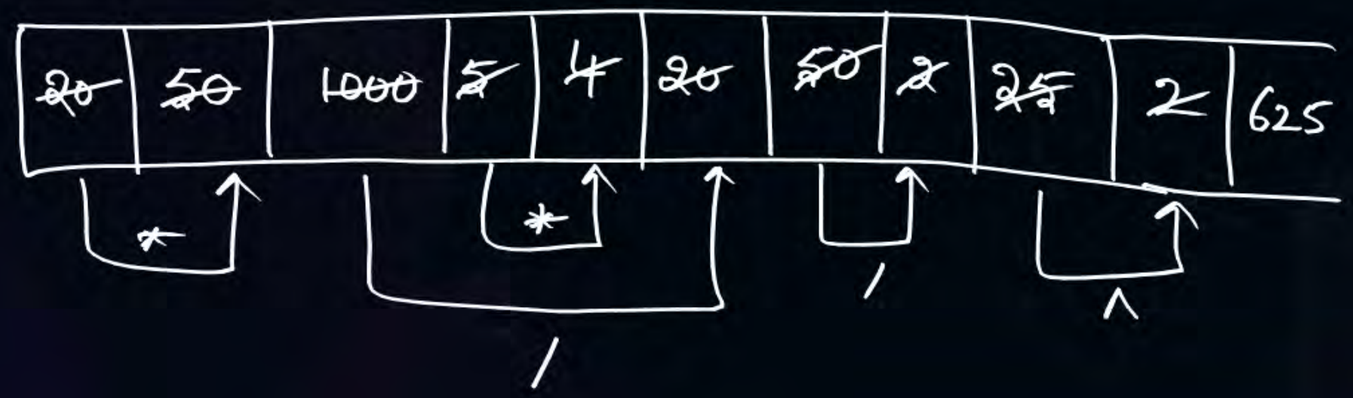
Topics to be Covered

– Practice Session

Topic : Practice Session - 1

#Q. The Result after evaluating the below Postfix Expression is 625 :

20 50 * 5 4 * / 2 / 2 ^



$= 25^2 = 625$

Topic : Practice Session – 1 GATE-2024-DA

#Q. Match the items in Column 1 with the items in Column 2 in the following table:

Column 1	Column 2
(p) First In First Out	(i) Stacks
(q) Lookup Operation	(ii) Queues
(r) Last In First Out	(iii) Hash Tables

(A)(p) – (ii), (q) – (iii), (r) – (i)
(B)(p) – (ii), (q) – (i), (r) – (iii)
(C)(p) – (i), (q) – (ii), (r) – (iii)
(D)(p) – (i), (q) – (iii), (r) – (ii)

Topic : Practice Session – 1– GATE-2024 - DA

#Q. The fundamental operations in a double-ended queue D are:

insertFirst(e) – Insert a new element e at the beginning of D.

insertLast(e) – Insert a new element e at the end of D.

removeFirst() – Remove and return the first element of D.

removeLast() – Remove and return the last element of D.

In an empty double-ended queue, the following operations are performed:

insertFirst(10)

insertLast(32)

a ←removeFirst() a←10 D [~~10~~ | 32]

insertLast(28) ✓

insertLast(17) ✓ D [~~32~~ | 28 | ~~17~~]

a ←removeFirst() a←32

a ← removeLast() a←17

The value of a is 17.

Topic : Practice Session – 1

#Q. . Suppose you are given an implementation of a queue of integers. The operations that can be performed on the queue are:

i. isEmpty (Q) — returns true if the queue is empty, false otherwise.

ii. delete (Q) — deletes the element at the front of the queue and returns its value.

iii. insert (Q, i) — inserts the integer i at the rear of the queue.

Consider the following function:

```
from collections import deque
def f(Q):
if not is_empty(Q):
        i = delete(Q)
        f(Q)
        insert(Q, i)
def is_empty(Q):    return len(Q) == 0
def delete(Q):    return Q.popleft()
def insert(Q, i):    Q.append(i)
```

What operation is performed by the above function f ?

(A) Leaves the queue Q unchanged

(B) Reverses the order of the elements in the queue Q

(C) Deletes the element at the front of the queue Q and inserts it at the rear keeping the other elements in the same order

(D) Empties the queue Q

Topic : Practice Session - 1

$f(S)$ $\boxed{\cancel{0}\ \cancel{-1}\ \cancel{2}\ \cancel{1}\ 3}$

#Q. A function f defined on stacks of integers satisfies the following properties.
$f(\phi) = 0$ and $\boxed{f(push(S, i)) = max(f(S), 0) + i}$ for all stacks S and integers i.

If a stack S contains the integers 2, –3, 2, –1, 2 in order from bottom to top, what is f(S)?

(A) 6 (B) 4 (C) 3 (D) 2

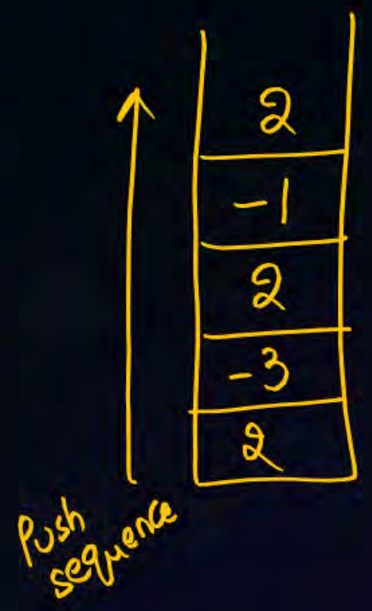
$f(Push(S, \overset{0}{\overset{i}{2}})) = max(0, 0) + 2 = 0 + 2 = 2$

$f(Push(S, -3)) = max(2, 0) + (-3) = 2 + (-3) = -1$

$f(Push(S, 2)) = max(-1, 0) + 2 = 0 + 2 = 2$

$f(Push(S, -1)) = max(2, 0) + (-1) = 2 + (-1) = 1$

$f(Push(S, 2)) = max(1, 0) + 2 = 1 + 2 = 3$

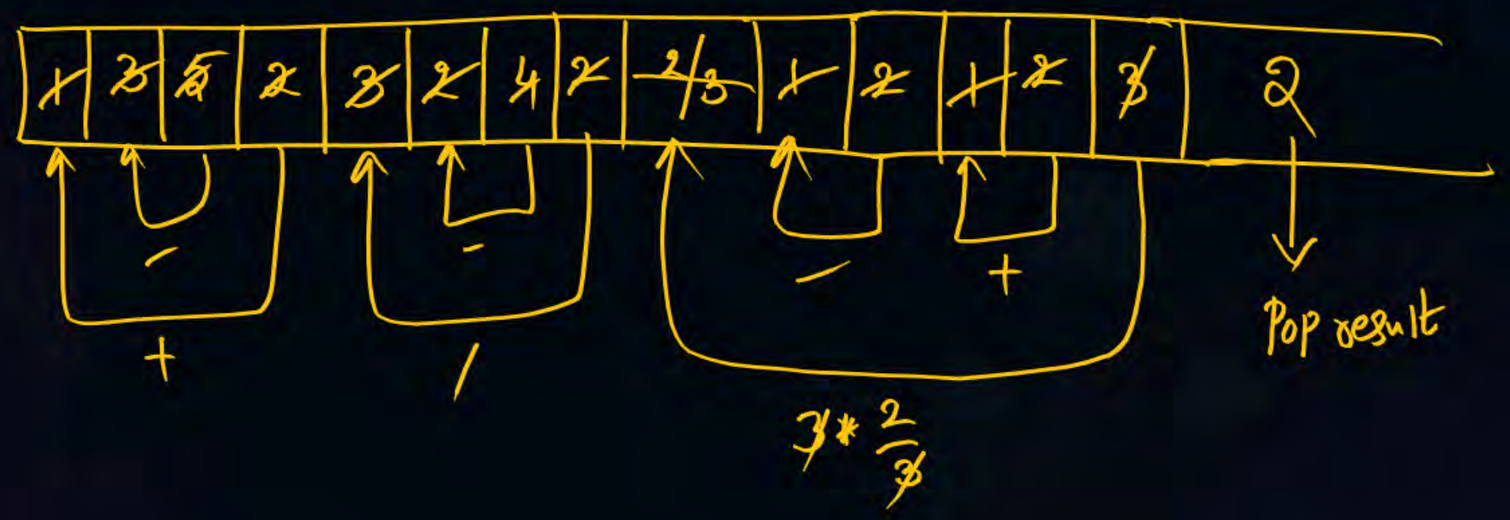


Topic : Practice Session – 1

#Q. Find the output of the following prefix expression

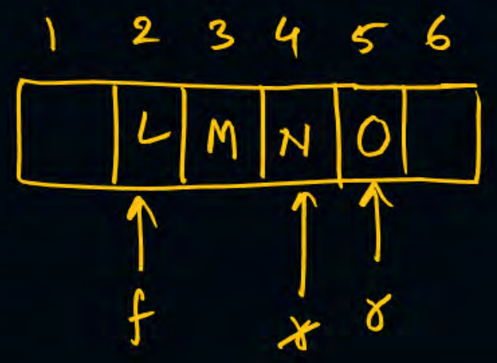
*+2-2 1/-4 2+-5 3 1

a) 2
b) 12
c) 0
d) 4



Topic : Practice Session – 1

#Q. Let the following circular queue can accommodate maximum six elements with the following data



front = 2 rear = 4, queue = _____; L, M, N, ___, ___

What will happen after ADD O operation takes place?

a) front = 2 rear = 5, queue = _____; L, M, N, O, ___

b) front = 3 rear = 5, queue = L, M, N, O, ___

c) front = 3 rear = 4, queue = _____; L, M, N, O, ___

d) front = 2 rear = 4, queue = L, M, N, O, ___

Topic : Practice Session – 1

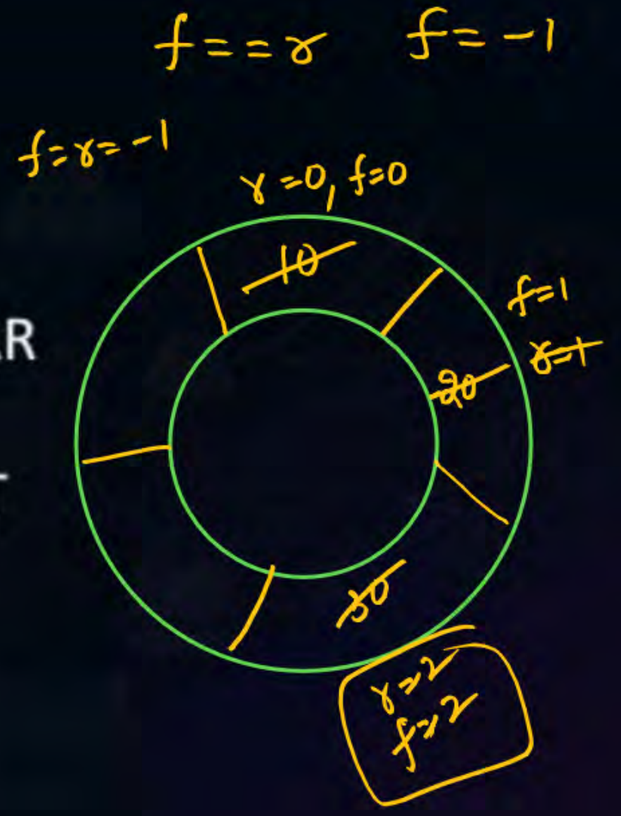
#Q. Suppose a circular queue of capacity (n) elements is implemented with an array(list) of n elements. Assume that the insertion and deletion operations are carried out using REAR and FRONT as index variables, respectively. Initially REAR=FRONT= -1. The conditions to detect queue full and queue is empty are?

a) Full: (REAR+1)mod n == FRONT Empty: REAR==FRONT ✓

b) Full: (REAR+1)mod n == FRONT Empty: (FRONT+1) mod n == REAR

c) Full: REAR==FRONT Empty: (REAR+1) mod n==FRONT

d) Full: (FRONT+1)mod n==REAR Empty: REAR==FRONT



'^' is right associative

X: a ^ b ^ c ^ d + e − f

More Priority → More

S	~~^~~	~~^~~	~~^~~	+	−
	Less Priority	Less		Pop	Pop
	Pop	Pop	Pop		

$\boxed{y: abcd\wedge\wedge\wedge e+f-}$

'^' right associative

X: a ^ b ^ c ^ d + e − f

Less more

Less more

more

S	−	+	^	^	^
				Pop	Pop

y: fedc^b^a^+−

Z: $\boxed{-+\wedge a\wedge b\wedge cdef}$

Q. Convert Infix Expression $a+b-c+d*e/f*g+h/i-j$

Less More Left Right More

Consider the Priorities and associativity as given below

Priority	Operator	Associativity
1	$-$	R TO L
2	$*$	L TO R
3	$+$	R TO L
4	$/$	R TO L

Less

S

+	/	+	*	/	*	+	/	-

Pop Pop Pop Pop Pop Pop

Y: abc-de*++fg*h+ij-//

Topic : Practice Session – 1

#Q. A simple queue is implemented using a list such that ENQUEUE and DEQUEUE operations are performed efficiently. Which one of the following statements is CORRECT (n refers to the number of items in the queue)?

(A) Both operations can be performed in O(1) time
(B) At most one operation can be performed in O(1) time but the worst case time for the other operation will be Ω(n)
(C) The worst case time complexity for both operations will be Ω(n)
(D) Worst case time complexity for both operations will be Ω(log n)

Topic : Practice Session – 1

H/W

#Q. The postfix form of the expression is (A+B) * (C*D-E)*F/G is

A. AB+CD*E-FG/**
B. AB+CD*E-*F*G/
C. AB+CD*E-F**G/
D. AB+CDE*-*F*G/

Topic : Practice Session – 1

#Q. Let ' s' be a stack and push and pop be functions implementing the Insertion and Deletion operations in a Stack. push takes 2 parameters: the stack and the element to be inserted , pop takes a single parameter: the stack. What will be the contents of the stack after the following operations: push(s,A), push(s,B), push(s,C), pop(s), pop(s), push(s, D), push (s, E), pop(s) ?

A. A D ✓
B. D E
C. C E
D. A E

Topic : Practice Session – 1

#Q. A program attempts to generate as many permutations as possible of the string, 'abcd' by pushing the characters a, b, c, d in the same order onto a stack, but it may pop off the top character at any time. Which one of the following strings CANNOT be generated using this program?

(A) abcd Push a Pop Push b Pop Push c Pop Push d Pop

(B) dcba Push a, b, c, d, Pop, Pop, Pop, Pop

(C) cbad Push a, b, c Pop, Pop, Pop, Push d, Pop

(D) cabd

THANK - YOU